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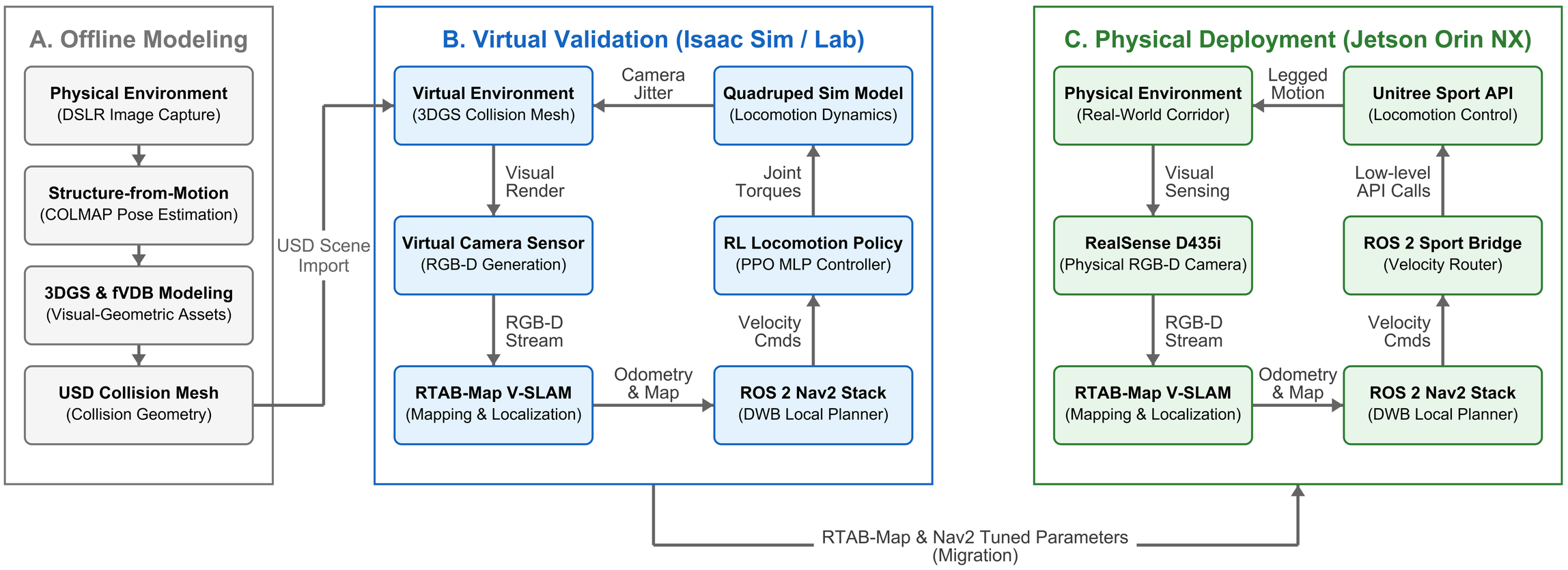
Computer Graphics and Vision Lab (CGV) · Handong Global University

Bridging the Sim-to-Real Gap in Quadrupedal Visual Navigation via 3D Gaussian Splatting

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Sim-to-Real Quadrupedal Visual Navigation Architecture

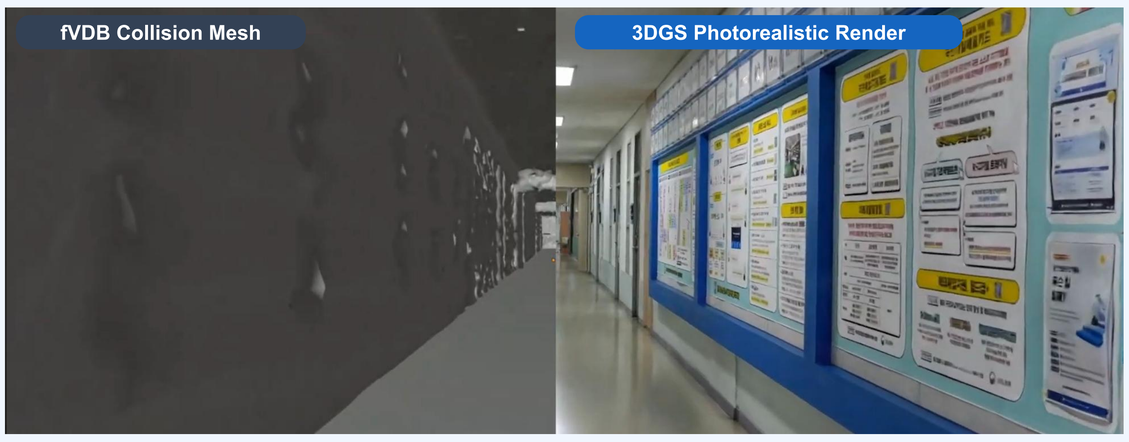


Key idea: RTAB-Map & Nav2 parameters tuned in the photorealistic 3DGS twin under RL gait bobbing (B) migrate directly to the physical Go2 (C) without retuning.

1. Motivation & Sim-to-Real Gap

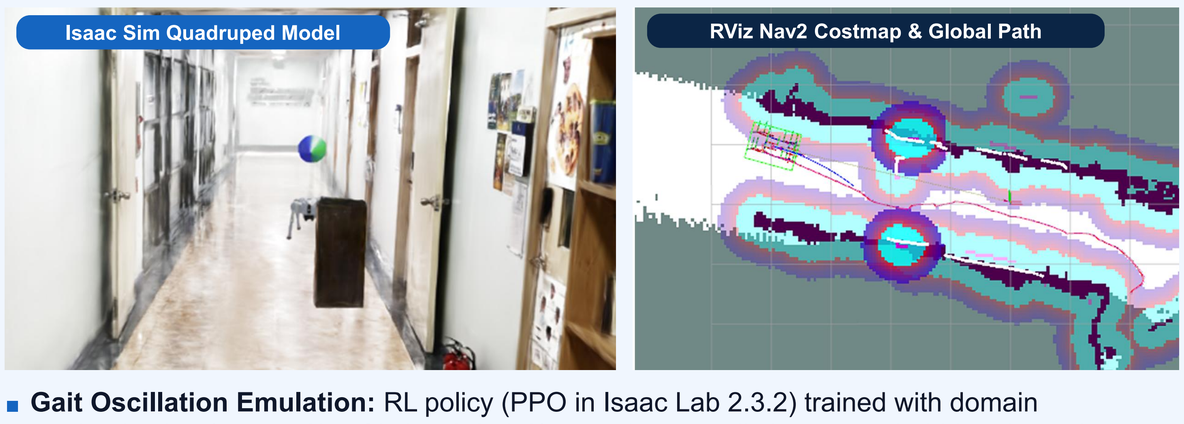
- **LiDAR-Free Quadruped Autonomy:** Single RGB-D camera replaces heavy, power-hungry LiDAR on legged robots.
- **Perceptual Sim-to-Real Gap:** Simulators lack photorealistic textures, causing V-SLAM feature tracking to fail in real environments.
- **Dynamic Gait Bobbing Noise:** Locomotion induces periodic camera jitter (pitch/roll/vertical), generating severe motion blur and false phantom obstacles.
- **Hardware Safety Bottleneck:** Physical trial-and-error tuning carries high collision risks and severe battery runtime limits.

2. 3DGS Digital Twin & Collision Mesh



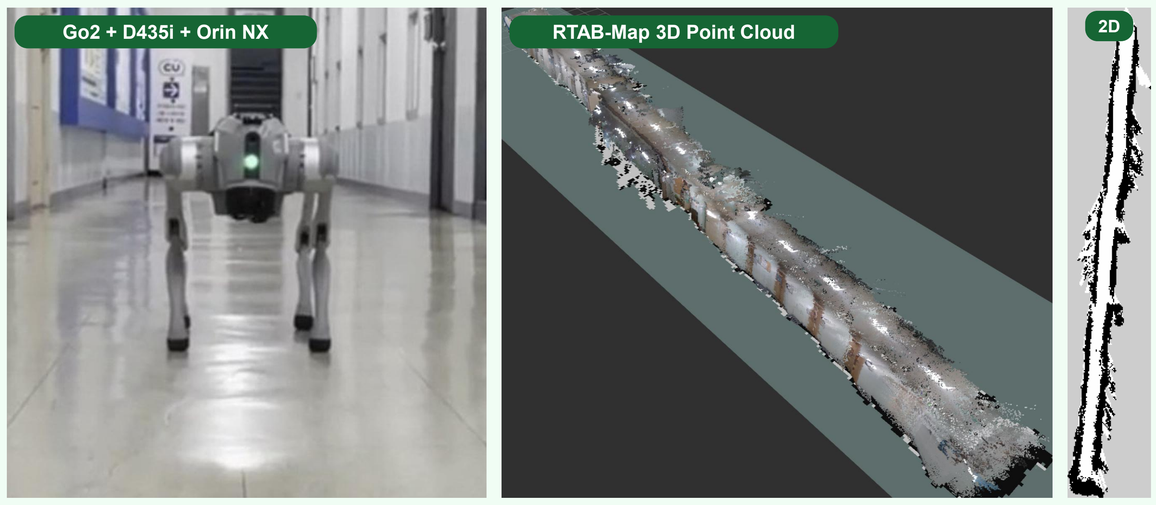
- **Offline Digital Twin Modeling:** Reconstructed from DSLR images via COLMAP Structure-from-Motion (SfM) and 3D Gaussian Splatting (3DGS) exported to USDZ format.
- **NVIDIA fVDB Sparse Volumetric Mesh:** Integrated into Isaac Sim 5.1.0 to compute accurate contact geometry and collision boundaries for quadruped foot placement.
- **Sensor Profile Alignment:** Virtual camera FOV, resolution (640x480), and optical center strictly match the physical Intel RealSense D435i sensor to eliminate hardware discrepancies.

3. Virtual Validation & Replanning



- **Gait Oscillation Emulation:** RL policy (PPO in Isaac Lab 2.3.2) trained with domain randomization (mass ±3kg, friction 0.6–0.8) reproduces realistic camera bobbing dynamics.
- **RTAB-Map V-SLAM Optimization:** Point cloud registration and loop closure thresholds tuned to withstand simulated locomotion pitch/roll oscillations without tracking loss.
- **Nav2 Local Replanning:** DWB local planner dynamically computes collision-free trajectories around inserted obstacles in the 80 m × 2.5 m digital corridor.

4. Physical Deployment (Unitree Go2)



- **Fully Onboard Edge Computing:** Intel RealSense D435i connected via USB 3.0 to an NVIDIA Jetson Orin NX, executing RTAB-Map and Nav2 autonomously in real time.
- **Decoupled Locomotion Control:** To conserve edge compute for SLAM, the heavy RL policy is bypassed on physical hardware; Nav2 velocity commands route directly to the built-in Sport API via a lightweight ROS 2 bridge.

5. Experimental Results & Verification

80 m 3DGS Corridor Digital Twin	**0.4 m/s** Walking Speed w/ Loop Closure
1 RGB-D Single D435i Camera (Zero LiDAR)	**0 Recalib** Zero-Code Sim-to-Real Transfer

- **Direct Parameter Migration:** Parameters optimized under simulated camera bobbing transferred seamlessly across ROS distributions (Jazzy on sim → Foxy on robot) without physical retuning.
- **Robust Visual Mapping:** RTAB-Map successfully extracted features and detected global loop closures throughout the 80 m corridor, eliminating phantom obstacles.

6. Conclusion & Future Outlook

- **Practical Virtual Sandbox:** Demonstrates that pairing a photorealistic 3DGS digital twin with RL-generated gait dynamics provides an effective, risk-free testbed for visual quadruped navigation.
- **Decoupled Edge Efficiency:** Separating simulation-based parameter discovery from onboard execution enables lightweight real-world deployment on resource-constrained Jetson platforms.
- **Future Research:** Expanding from static corridors to dynamic obstacle avoidance benchmarks, multi-floor terrain transitions, and direct neural policy deployment.

References: [1] B. Kerbl et al., "3D Gaussian splatting for real-time radiance field rendering," *ACM TOG*, 2023. [2] M. Labbé & F. Michaud, "RTAB-Map as an open-source lidar and visual SLAM library," *arXiv:2403.06341*, 2024. [3] S. Macenski et al., "Robot Operating System 2: Design and architecture," *Science Robotics*, 2022. [4] S. Zhu et al., "VR-Robo: Real-to-sim-to-real for visual robot navigation," *IEEE RA-L*, 2025.